

Erosion of Seasonality Anomalies in US Equity Markets

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Introduction

As Halloween approaches, traders and investors are focusing on the market's ups and downs. However, not many people are aware of the existence of a Halloween strategy, also known as the Halloween effect. The Halloween effect traces its origins back to 16th-century London, England. During that time, the wealthy class would leave the city to spend the summer at their country estates, often neglecting their investment portfolios, and return in September. This practice has evolved over the past two centuries into the saying: "Sell in May and go away." In this report, we reference the publication "[Evidence of Alpha Decay in US Equities Markets](#)" by the CUHK Quant Trading Society, exploring seasonality analysis and backtesting trading strategies in greater depth. Our primary goal is to investigate the seasonality of the US stock market and discuss the decline in the profitability of seasonal investment strategies.

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1 Data

The S&P 500 data serves as a proxy for the US equity market. Data from 1980 to 2023 were downloaded using the yfinance API, and we utilized the adjusted close values for each trading day in this report.

2 Seasonality Analysis

2.1 Monthly Seasonality

The Halloween strategy posits that equity returns from November through May exceed those from June through October, while the “January effect” refers to the historically higher average return in January. As shown in Figure 1, January does indeed post above-average but not superior return, since May, November and December exhibit similarly strong performance. Moreover, the premium has diminished in recent years, indicating that the January effect may be fading.

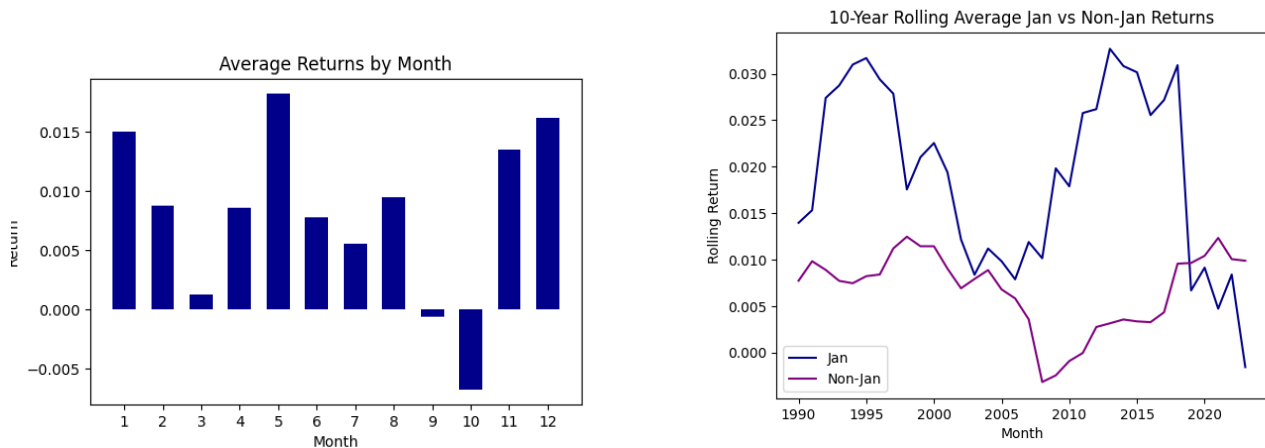


Figure 1: Monthly Graphs

To investigate monthly seasonality more formally, we apply a simple additive decomposition:

$$Y_t = T_t + S_t + \varepsilon_t$$

, where $Y_t, T_t, S_t, \varepsilon_t$ denote the observed return, trend, seasonal and residual components, respectively.

The estimated seasonal factors plotted in Figure 2 lend support to the Halloween strategy. The seasonal component is negative from June through October and turns positive from November through May. In particular, November, December, January and May show the largest positive seasonal lifts, whereas the June–October interval forms a clear seasonal trough.

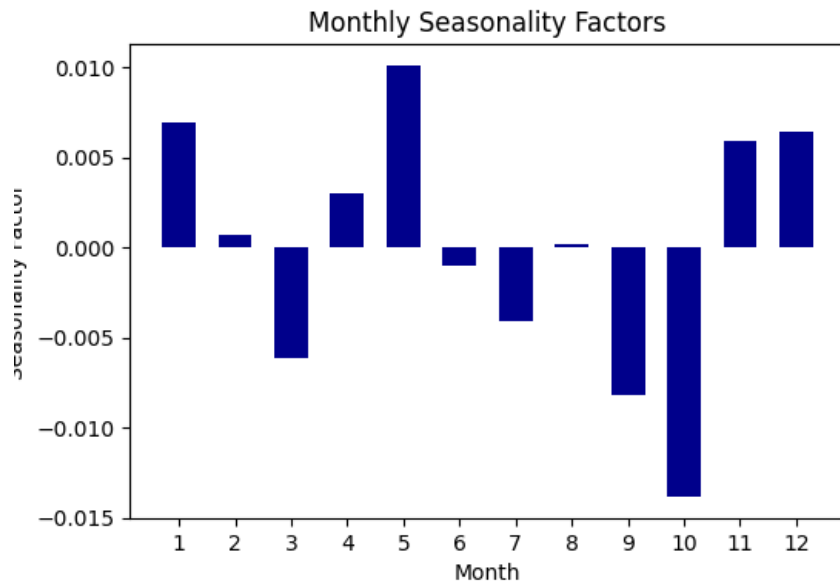


Figure 2: Monthly Seasonality Graph

2.2 Daily Seasonality

Beyond monthly effects, daily seasonality also emerges in equity returns. The Turn-of-Month effect hypothesis states that returns are abnormally high around the turn of the calendar month. Empirically from figure 3, we observe extraordinary average returns on the last five trading days of the outgoing month and the first three trading days of the new month.

When we decompose the daily return series, the seasonal component indeed exhibits a pronounced positive peak on the final trading day of each month. By contrast, the seasonal components for the first three trading days do not rise consistently.

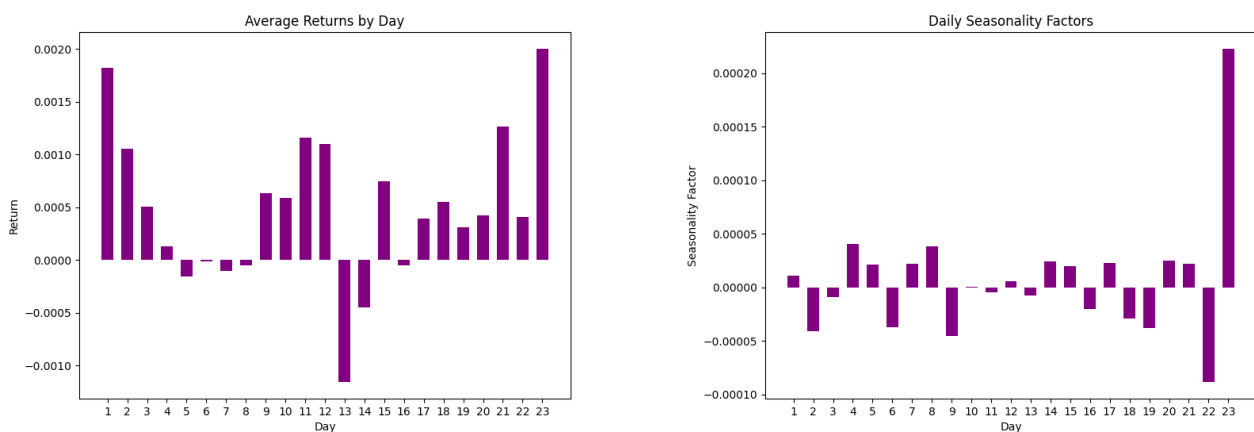


Figure 3: Daily Seasonality Graphs

2.3 January Barometer

The January Barometer asserts that January's stock-market return predicts the performance for the remainder of the year. However, it rests on little theoretical foundation and, as our tests show, it does not hold up empirically. We computed both Pearson's and Kendall's τ correlation between January returns and annual returns. The correlation coefficient is -0.438 in years when January returns are negative and -0.084 when January returns are positive. In addition, the unstable, near-random correlations that appear in Figure 4 indicate no reliable predictive link. Hence, we reject the January Barometer as a meaningful seasonal effect.

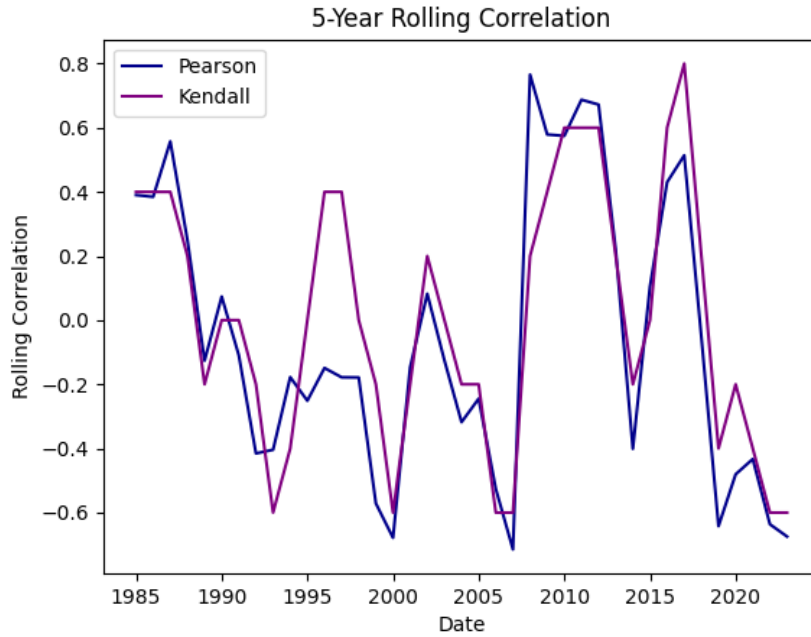


Figure 4: Jan Correlation Graph

3 Trading Strategies

3.1 Monthly Seasonality Strategy

Our monthly seasonality analysis reveals a distinct trough from June to October, along with a negative seasonal component in March. Consequently, we backtest a strategy where:

1. Buy at the close of the first trading day in November and sell at the close of the last day in February
2. Buy at the close of the first trading day in April and sell at the close of the last day in May

3.2 Monthly Seasonality Strategy + Turn-of-Month Effect

Our daily seasonality analysis offers some support for the Turn-of-Month effect. Therefore, we implement the following actions in addition to our monthly seasonality strategy:

1. Buy at the close of the 19th trading day each month and sell at the close of the 3rd trading day of the following month
2. If there are buy signals from both the monthly seasonality strategy and Turn-of-Month effect, we will apply leverage of 50%.

4 Backtest Result & Discussion

4.1 Backtest Result

Our backtest spans from 1980 to 2023, applying a 10 bps transaction cost and no leverage costs for simplicity:

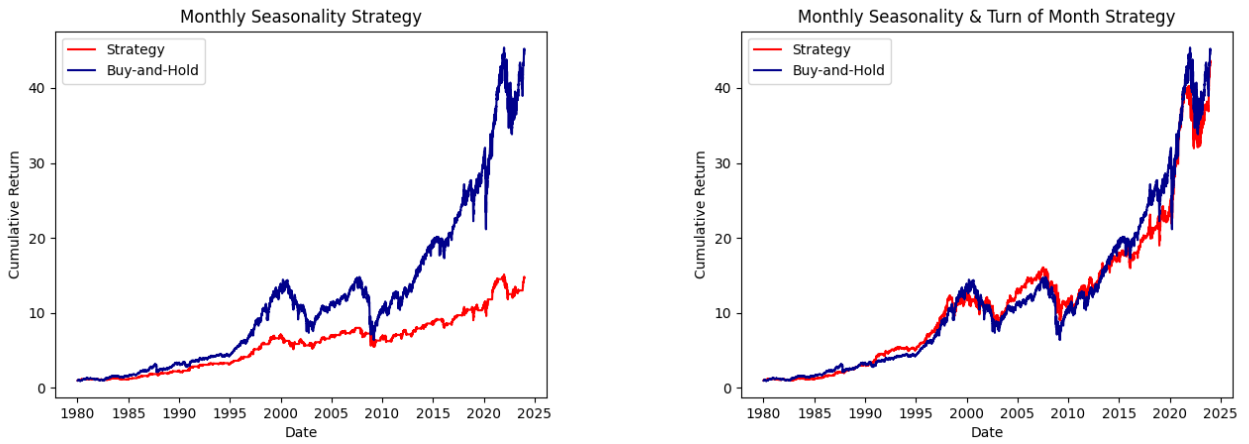


Figure 5: Portfolio Performance Graphs

Strategy	Ann. Return (%)	Ann. Volatility (%)	Sharpe Ratio	Sortino Ratio	Max Drawdown
1	6.31	11.6	0.371	5.55	-0.315
2	9.20	15.7	0.458	8.01	-0.426
3	9.04	17.9	0.392	7.85	-0.568

Table 1: 1: Monthly Seasonality; 2: Monthly Seasonality + Turn-of-Month Effect; 3: Buy-and-Hold

The stand-alone Monthly Seasonality Strategy clearly underperforms as it delivers the lowest annualized return and the worst Sharpe ratio of the three. By contrast, the Monthly Seasonality Strategy enhanced with the Turn-of-Month effect not only edges out buy-and-hold in outright return, but does so with appreciably less risk. It yields lower volatility and max drawdown. As a result, this strategy achieves the highest Sharpe and Sortino ratios.

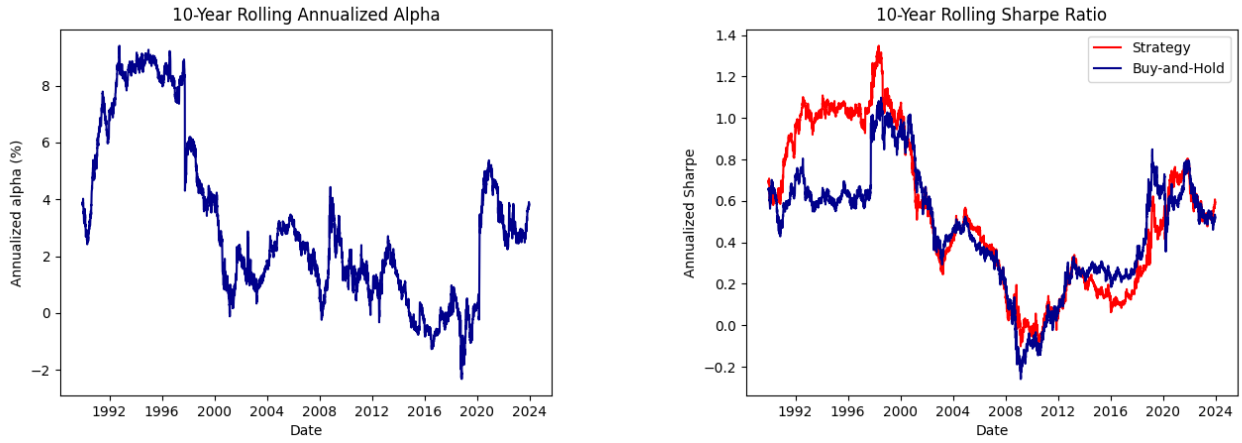


Figure 6: Rolling Portfolio Performance Graphs

4.2 Discussion

In a deeper dive on “Monthly Seasonality Strategy + Turn-of-Month Effect“, we observe a clear erosion of profitability over time. As shown in Figure 6, its 5-year rolling alpha over the buy-and-hold benchmark exceeded 4% annually in the pre-2000 era, but has plunged into the 0–2% range from 2000 to 2020. A very similar pattern appears in the strategy’s rolling Sharpe ratio. It demonstrated a substantial lead over buy-and-hold before 2000, yet since the turn of the millennium their Sharpe profiles have moved almost in tandem.

The decay might be due to technological advancement. Today, virtually every data vendor offers real-time or intraday price feeds, corporate calendars, and economic releases, whereas a hedge fund two decades ago might have waited days to receive cleaned data. Coupled with cloud services and affordable GPU clusters, researchers can now spin up thousands of backtests in parallel, calibrating dozens of model variants in hours rather than weeks.

Meanwhile, trading infrastructure has transformed beyond recognition. Electronic trading platforms and direct API access to exchanges have driven both commissions and latency to historic lows. A buy-and-hold fund in the 1990s might have paid 1% per trade and waited minutes for a fill, while retail platforms today often charge zero commissions and execute orders in milliseconds. With the frictions all but gone, what used to be a niche seasonal anomaly is now widely deployed, so any fleeting edge is quickly arbitrated away.

4.3 Moving Forwards

Our analysis has focused exclusively on the S&P500, which likely understates the full extent of calendar-based anomalies. Small- and mid-cap stocks, as well as many international markets, may exhibit significant seasonality due to lower liquidity, narrower analyst coverage, and higher access barriers.

On the modelling side, our seasonal decomposition might be too naive that a more robust framework, such as a Bayesian Structural Time Series model, could estimate the seasonality fac-

tor more accurately. Furthermore, incorporating variables such as aggregate fund flows, liquidity cycle indicators, shifts in investor sentiment, tax-loss-harvesting windows, monetary policy transitions and recessionary episodes into a multivariate setup could identify hidden patterns beyond seasonality.

As for implementation, we could move on with developing a realistic model that layers in bid-ask spreads, slippage, market-impact estimates, financing costs for leverage and transactions.